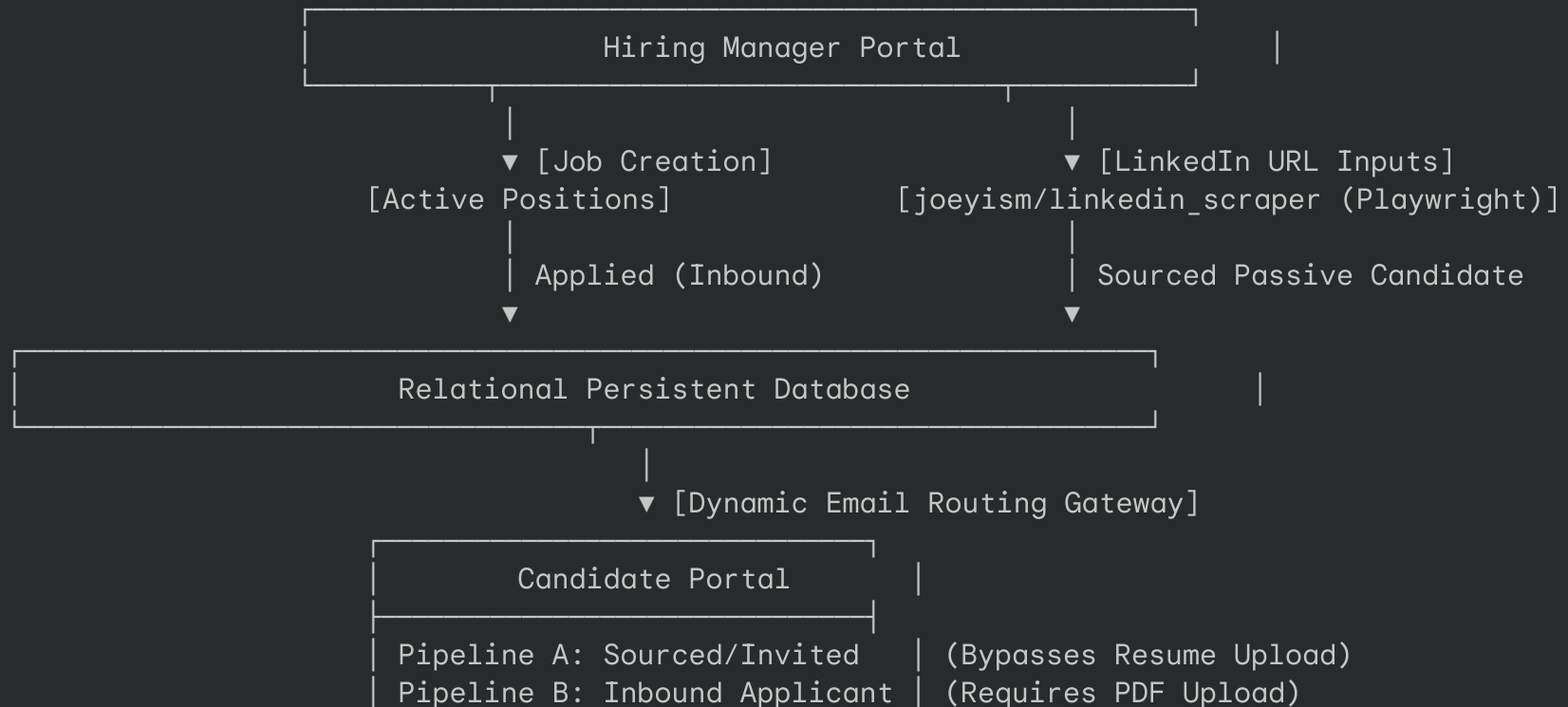


Architecture & Agent Specification: Hybrid Dual-Portal Workspace with direct Browser Scraper & Automated Sourcing

This document establishes the official technical specifications, multi-agent orchestrations, and user interface mappings for the single-company, dual-portal "Intelligent Recruiter Workspace" using direct browser automation.

🔧 Unified System Topology & Dynamic Routing

The system manages a company's internal talent pipeline, offering two entry portals powered by a 6-Agent AI Core.





1. The Candidate Portal Specification (The Warm-up Sandbox)

An interactive candidate workspace designed to replace the standard cold technical assessment with a welcoming, collaborative coding or architectural "warm-up sandbox."

Interface & Interactive Elements:

- **Dynamic Authentication Widget:**

- `st.text_input("Enter Email Address")` : The primary user session lookup field.

- **Router Engine Logic:**

- **IF Candidate is Invited/Pre-Sourced:**

- Renders a high-visibility welcome banner: *"Welcome back, [Candidate Name]! You have been hand-selected for our [Position Name] role based on your LinkedIn achievements."*
- Bypasses the PDF resume uploader interface.
- Renders a status progress bar marked at **50% (Sourced & Invited)**.
- Displays the personalized recruitment email drafted by the **Report Agent** in an expandable info box.
- Unlocks the **Interactive Sandbox Screen** immediately.

- **IF Candidate is an Inbound Applicant:**

- Renders a standard application interface.
- `st.selectbox("Select Open Role")` : Populates dynamically from active roles created by the hiring manager.
- `st.file_uploader("Upload Resume (PDF format)")` : Collects the unstructured candidate profile.
- `st.button("Submit My Application")` :

2. Invokes the **Resume Agent** to run raw standardization and prestige neutralization.
3. Invokes the **Matching Agent** to execute the Critical Recruiter vs Talent Advocate debate.
4. Invokes the **Candidate Interview Agent (Phase A)** to pre-generate 3 targeted warm-up questions addressing their specific profile gaps.
5. Saves applicant data to the relational database with status set to `"Applied"` and updates progress tracker to **33%**.

- **The Interactive Sandbox Screen:**

- Accessible to both Sourced and Inbound candidates who have unlocked their screening.
- Renders 3 highly-personalized custom questions inside individual `st.text_area` input fields.
- `st.button("Submit Sandbox Answers")` :
 - Validates input length (minimum 10 characters per question).
 - Invokes the **Candidate Interview Agent (Phase B - Evaluation)** to score responses from 0 to 100.
 - Invokes the **Report Agent** to generate a custom 3-week upskilling roadmap.
 - Updates status to `"Screening Completed"` and sets progress tracker to **100%**.
 - Renders the "Value-Back" Feedback layout, displaying immediate qualitative coaching and training milestones directly to the candidate (even if their score falls below the hiring threshold).



2. The Hiring Manager Portal Specification

The main workspace where recruiters manage jobs, trigger the direct `linkedin_scraper` library via Playwright, review de-biased candidate lists, and analyze candidate trajectory.

Interface & Interactive Elements:

- **Job Specification Builder Workspace:**

Description").

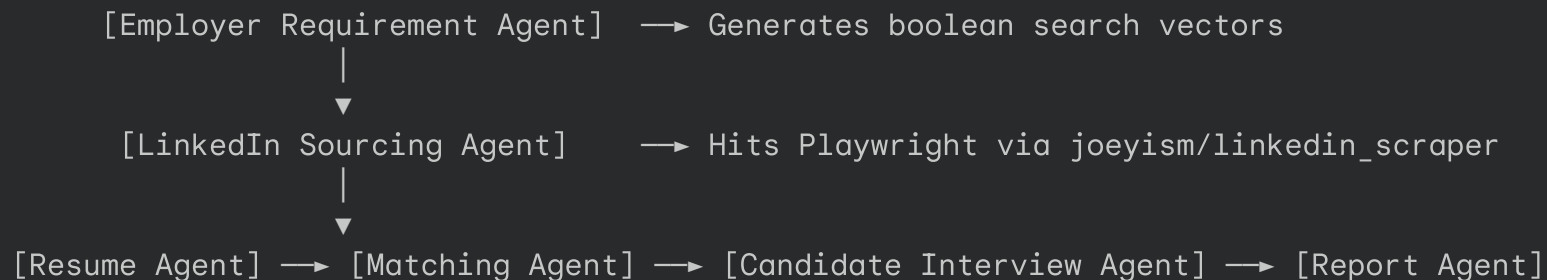
- `st.button("Publish Job Posting") :`
 - Calls the **Employer Requirement Agent** to extract job pillars and construct search query arguments.
 - Appends the new job object to the relational database dictionary.
- **Direct Browser LinkedIn Scraper Console:**
 - `st.selectbox("Target Position to Align With") :` Associates scraped profiles with published roles.
 - `st.text_input("Public LinkedIn Profile URL to Scrape") :` URL input targeting the browser automation parser.
 - `st.button("Fetch, Scrape & Analyze Profile") :`
 - Launches an asynchronous Playwright browser instance via `linkedin_scraper`'s `BrowserManager`.
 - Loads the authenticated session state from the credentials panel or saves a new browser session locally.
 - Executes `PersonScraper` to extract `name`, `headline`, `location`, `about`, `experiences`, and `educations`.
 - Serializes retrieved profile structures natively into target schemas using Pydantic schema utilities.
 - Invokes the **Matching Agent** to output a baseline similarity matrix.
 - Invokes the **Candidate Interview Agent (Phase A)** to pre-generate 3 targeted screening questions.
 - Invokes the **Report Agent** to write a personalized outreach email template and draft a "Why This Person?" recruiter pitch.
 - Stores the results in a temporary staging state, displaying live pipeline logs in `st.code`.
- **Sourced Profile Staging Card:**
 - Visualizes retrieved LinkedIn metrics, showing key match results, pros/cons, and the custom email draft.
 - `st.button("Send Invitation & Unlock Interactive Screening") :`
 - Saves the staged candidate to the database with status set to `"Invited"`.

- **Active Candidate Pipeline Dashboard:**

- `st.toggle("Activate Prestige Neutralizer Mode")` : Replacing company and university brand names (e.g. Google, Harvard) with descriptive categories (e.g. `[Tier-1 Tech Corporation]`, `[Tier-1 Research University]`) to force a focus on pure execution, skills, and trajectory.
- `st.plotly_chart()` : Generates a 2D Scatter Plot mapping candidates along **Trajectory Slope (Learning Rate)** vs. **Core Match Fit (%)**, exposing highly capable but unpolished "hidden gems" in the top-left quadrant.
- `st.expander("Detailed Evaluation Panel")` for each candidate:
 - Displays structured side-by-side comparative notes (Advocate Pros vs. Recruiter Cons).
 - Displays candidate screening answers paired with the **Evaluation Agent's** automated grading and 3-week upskilling roadmaps.

3. AI Multi-Agent Layer Specifications

The system utilizes 6 specialized agents orchestrating targeted prompts to process, score, search, and guide applicants.



A. Resume Agent (Standardizer & Prestige Neutralizer)

- **Execution Parameters:** Model: `gpt-4o-mini` | Temperature: `0.1`

clean JSON. When requested, replaces prominent institutional names (e.g. Yale, Google, McKinsey) with descriptive, standardized classifications (e.g. [Tier 1 Ivy League School], [Tier 1 Consulting Firm]) to prevent systemic prestige bias.

B. Employer Requirement Agent (Profiler)

- **Execution Parameters:** Model: `gpt-4o-mini` | Temperature: `0.2`
- **Duty:** Analyzes job descriptions, extracts primary technical and behavioral pillars, and drafts optimized Boolean query vectors designed to search LinkedIn directories.

C. LinkedIn Sourcing Agent (Direct Browser Scraper)

- **Execution Parameters:** Playwright Core async worker driving `joeyism/linkedin_scraper` models.
- **Duty:** Automates standard user interactions using loaded browser sessions. Navigates, expands content nodes, and converts native CSS components into a clean structured Pydantic record. Uses custom exception fallbacks for headless browser constraints.

D. Matching Agent (Hiring Committee Debate)

- **Execution Parameters:** Model: `gpt-4o-mini` | Temperature: `0.4`
- **Duty:** Simulates a hiring committee debate between a *Critical Recruiter* (spotting gaps, tenure risks, and lack of requirements) and a *Talent Advocate* (finding core potential and transferable skills). Computes standard numerical metrics across technical, domain, and culture categories.
- **Trajectory Core Calculation:** Evaluates velocity of growth (e.g. moving from Junior Developer to Tech Lead in under 2 years, or building high-complexity personal repos without formal academic credentials) and generates an explicit **Trajectory Slope Score (0-100)** representing raw potential.

E. Candidate Interview Agent (Screening & Evaluation)

- **Execution Parameters:** Model: `gpt-4o-mini` | Temperature: `0.3` to `0.5`

probe the candidate's unique profile gaps.

- *Phase B (Evaluation)*: Analyzes submitted answers against the target job requirements, returns a final grading score (0-100), and provides actionable, qualitative performance feedback.



F. Report Agent (Synthesizer)

- **Execution Parameters:** Model: `gpt-4o-mini` | Temperature: `0.3`